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# A comparative ethological study of Nilgai (*Boselaphus tragocamelus*) in the Gopalganj District of Bihar

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**Abstract-** Nilgai is the largest Antelope and due to its crop raiding behaviour causing huge economic loss, it has been declared vermin and has been ordered by many States Government for culling. The culling proposal has already been implemented in Bihar. The present paper deals with the study conducted on the various behaviour of Nilgai like Aggressive behaviour, social behaviour, Reproductive behaviour, Defence behaviour, Docile behaviour etc. by using direct sighting and Camera Photography and Questionnaire cum Survey method and direct conversation with the local inhabitants, and focussing their awareness about its future domestication prospects if possible and its protection as it is the sole member of its genus. During the study it was seen that the local village population was very much less aware about its docile and reproductive behaviour rather they were very much acknowledged about the cases of its Aggressive and Crop-raiding behaviour. Still, they emphasized the protection of their agricultural crops by planting Lemon-grass, Cactus, Sunflower along the margins of their field.

**Key words: Antelope, Awareness, Crop-raiding, Docile, Vermin**

## INTRODUCTION

The Conservation International had identified 17 megadiverse countries in 1998 and all of them exhibit great biodiversity and there is endemism at the level of species, genera, and families. India is one of the members of group of Like – Minded megadiverse countries. India is having biodiversity index of 301.63, encompassing about 400 mammalian species. There are 31 species of ungulates found, of which 25 are protected in the wildlife protection act 1972. There are 6 antelope species namely Nilgai, four horned Antelope, Indian Gazelle, Blackbuck, Tibetan Antelope, Tibetan gazelle found in India. Out of these 3 are endemic to the Indian subcontinent. the Nilgai or Blue-bull is one of the largest Antelope in the Indian subcontinents.[1] The gross feature of Nilgai includes longer forelegs often marked with white socks thus called white footed antelope. It belongs to family *Bovidae*, sub-family *Bovinae* and tribe *Boselaphini*. It is the only species of *Boselaphus*.[2] It is distributed in India from Himalayan foothills southward through Central India to Mysore.[3] It is a browser or mixed feeder but primarily it is a grazer. It prefers paddy, maize and wheat crops.[4] It is categorized

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as least concerned animal by IUCN red list.[5] In India Nilgai is protected under the schedule III of the wild life protection act 1972. Due to increased human population growth and Industrialization the natural Habitat of Nilgai has been depleted and so is the case of their migration towards human settlement periphery.[6] Moreover, their adaptive ability is very high for new Habitat which is also the reason for their increasing population.[7] They are migrating in search of food towards cultivated grass, agricultural crops and periphery of village area.[8]

It causes economic loss by crop-raiding, so it has been declared as vermin and has been ordered by many state governments for culling. Some districts of Bihar where maximum crop damage has been reported are Kaimur, Buxar, Bhojpur, Vaishaili, Motihari, Supaul, and Nalanda and Patna.[9] The culling proposal has already been implemented in Bihar.[10]

## MATERIALS & METHOD

**Study area:** The Gopalganj district is a part of Saran Division and is located in the western north corner of Bihar state covering an area of 2033 KM$^2$. (fig 1) Geographically it is located between 83.54° - 85.56° latitude and 26.12° - 26.39° north longitude. The altitude is 73 feet above MSL. The soil of the district is thick Alluvium deposited by River Gandak. The pH of the soil is mostly alkaline ranges from 7-8. The temperature ranges from 4°C. in winter (Dec. to Jan.) and more 44°C. in hot summer (May to June). The average annual rainfall is about 1218 mm. But there is large variation in rainfall over year to year. It is bounded on east by Champaran and River Gandak, on the south by Siwan district and on the north-west Dewaria district of U.P. The river Gandak is supported by tributaries like Khanwa, Jharhi, Daha, Dhanahi etc. Due to this the land of district is fertile and alluvial. So, cultivation and irrigation prospects are good. The River Gandak by depositing the top quality of soil brought from Nepal and plays an important role in the economy of Gopalganj district. This district is an important agricultural centre in north Bihar and has several secondary and tertiary industries based on agriculture. This district is one of the largest sugarcanes growing district in India and also known for its advances in horticulture.

The methodology followed was direct sighting of animals from a close distance and observing the behaviour of animals by using photography and then analysing the observations.[11] A questionnaire survey was carried out to know the awareness of village people about the various behaviour of nilgai like reproductive behaviour, aggressive behaviour, social and docile behaviour etc. Moreover, direct conversation with local villagers was done. A total of 130 people were interviewed in which preference was

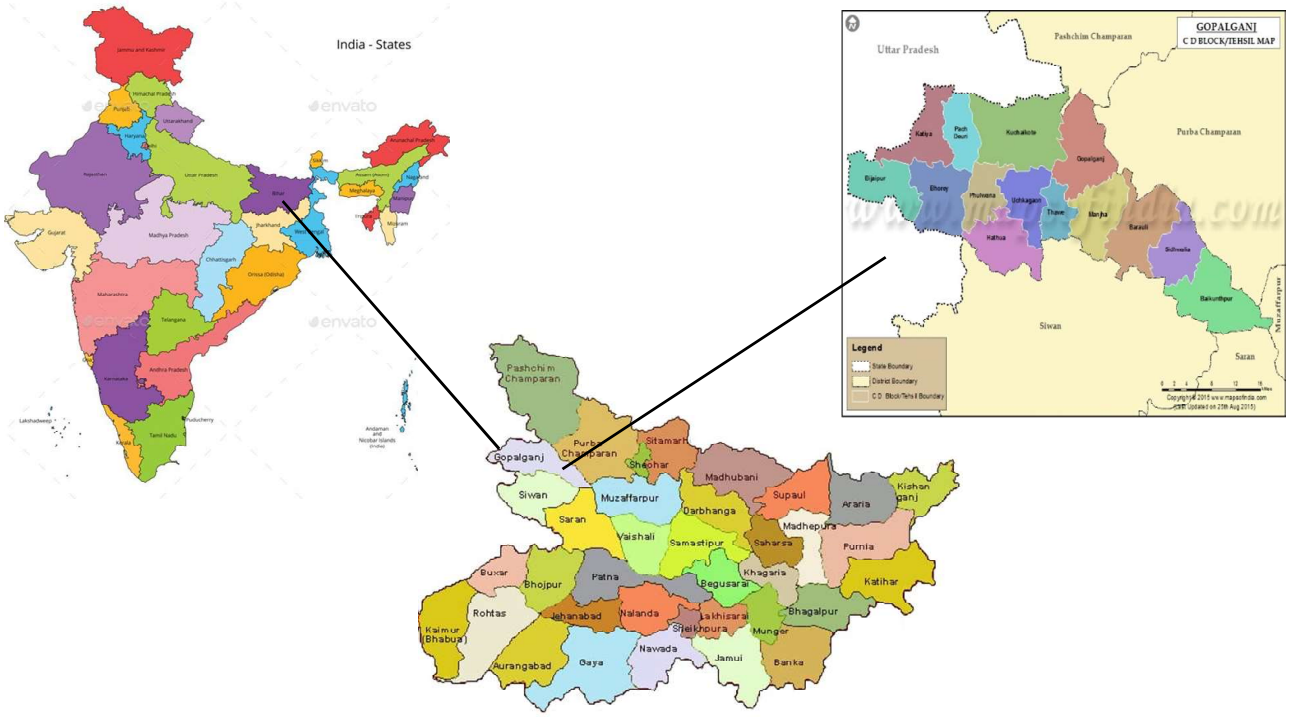



**Fig 1. Showing the study area in the Gopalganj district of Bihar**

given to the farmers working in the field where the animal was observed. The selected areas were Balesara and Sabeya village of Uchkagaon block and Sasamusa block of Gopalganj (fig1). This study was done in the above mentioned region from October 2022 to September 2023.

## RESULTS & DISCUSSION

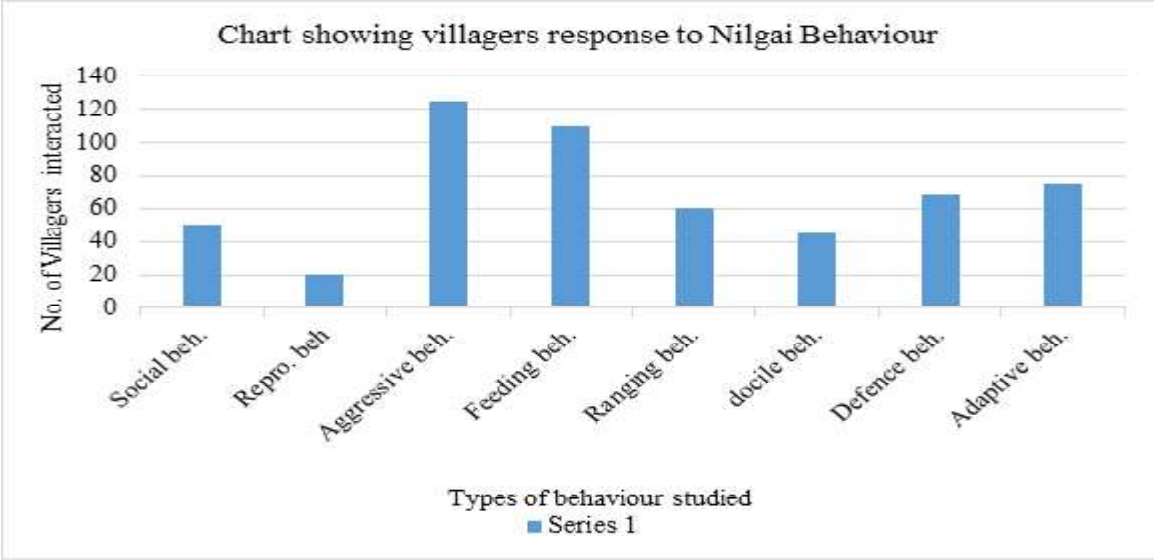
After the direct conversation with the villagers and analysis of questionnaire cum survey form in which a frame of questions related with behaviour of the Nilgai was asked and also the method to protect the crops was emphasised, the following inferences were made (Graph 1):

**I. Social behaviour**: Nilgai are partially social in nature. They follow sexual segregation except during the breeding season (December to March). The males form a separate herd. They were almost diurnal and sometimes nocturnal in habit. Large groups were rarely seen.

The group size of the animal during the study were as follows:-

- 08 Adult females with Calves
- Male groups varying in number from 3-7 in the Balesara village.
- It was seen that sometimes mixed group was seen in 12:3 ratio during winter in which both male and female were seen and the number of females was more.

**II. Reproductive Behaviour**: The villagers were having poor knowledge about the reproductive behaviour. They said that it reproduces twice in a year but observation was different. Male and female were shown clear sexual dimorphism and the mating season was in winter from December to March and gestation period was 8 to 9 months and mostly twin calves are born.




**Graph 1- Column chart showing number of villagers responded to questions regarding Nilgai behaviour**



**Fig. 2- Bull along with female Nilgai in the Balesara region of Gopalganj**

A herd of female was seen grazing with bull and meanwhile the bull mated with one of the females. This behaviour was confirmed by the villagers also. So, it may be a sign of interbreeding. (fig. 2)

**III. Aggressive behaviour**: The villagers narrated many cases of aggressive behaviour of Nilgai. They often caused accidents in the Highway after collision with vehicles. They are afraid of light and sound of the vehicle specially during the night time and when the Nilgai is roaming alone directionless. Between male and female, it was observed that male was more aggressive.

**IV. Feeding behaviour/Crop raiding behaviour**: The Nilgai was seen grazing as well as browsing in the field. They can thrive on variable proportion of grass herbs. They also prefer to eat paddy, maize, gram and sugarcane. They damage the crops by feeding and trampling. The villagers were very disturbed economically by the crop raiding behaviour. Nilgai destroys sugarcane plantation most in the area.

**V. Ranging behaviour**: The ranging pattern have already been studied but very little is known about the movement pattern of Nilgai.[12] According to the villagers the animal prefers its habitat where availability of fresh vegetation is more, basically they prefer to dwell near the agricultural fields and most notable feature of this area is kusha grass scientifically known as *Desmotachya bipinnata* which is an ideal place for the hiding of the Nilgai.

**VI. Docile behaviour**: The animal was seen grazing with cattle especially in the Balesara and Sabeya village. It was found very comfortable with Cattle egret (*Bubulcus*) as well as Mynah bird (*Acridotheres tristis*) was sitting on the animal's back (fig. 2 & 3). It was observed that Nilgai herd was seen along with local dog's group. Even the dog barked on the Nilgai herd, but the animal remained non respondent, busy in grazing. This is a sign towards its possible domestication in near future.

We can harness their docile behaviour for future domestication prospects for improving livelihood of local rural population.[13,14]

**VII. Stress tolerant/Adaptive behaviour**: The Nilgai was seen grazing in the field even during the midnoon of June when the temperature of the area reached at about 44°C. Meanwhile no other cattle or animal was seen in the field.

**VIII. Defense behaviour**: – It was observed that when the animal in alone was active it prefers green grasses but while resting it prefers to rest in brown dry grass but maybe it points towards their defensive mechanism (fig. 4), as when the animal was seen in herds they were seen resting in green grasses also. while resting also some of the members used to stand and other members were resting, might be for signalling a coming danger at least two or three members always stand in a group and those members are generally the elder adult member of the group and they always face the side of entrance of the field. (fig. 5)



**Fig. 3- Herd of Nilgai grazing with cattle in the Mirganj**



**Fig. 4- A Nilgai resting in dry grasses in Sabeya block of Gopalganj**



**Fig. 5- Herd of Nilgai resting in green grass**

During direct conversation, the villagers said that they protected their field by growing cactus, sunflower or in some region's lemongrass along the margin of the field also which repels the Nilgai.

## CONCLUSION

There is very less awareness about the social, docile behaviour and reproductive behaviour of Nilgai but their aggressive and crop-raiding behaviour is much highlighted. So, we have to study the ethological aspect of the animal and measures to protect the Nilgai by using some innovative measures as it is the sole member of its genus. We need to explore their ecological function and importance. and emphasize on the docile behaviour of Nilgai for its future domestication if possible leading to upliftment of economic condition of rural mass

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